```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("car data (1).csv")
df.head()
```

	Car_Name	Year	Selling_Price	Present_Price	Driven_kms	Fuel_Type	Selling_type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Car_Name       301 non-null    object
 1   Year           301 non-null    int64
 2   Selling_Price  301 non-null    float64
 3   Present_Price  301 non-null    float64
 4   Driven_kms     301 non-null    int64
 5   Fuel_Type      301 non-null    object
 6   Selling_type   301 non-null    object
 7   Transmission   301 non-null    object
 8   Owner          301 non-null    int64
dtypes: float64(2), int64(3), object(4)
memory usage: 21.3+ KB
```

```
df.describe()
```

	Year	Selling_Price	Present_Price	Driven_kms	Owner
count	301.000000	301.000000	301.000000	301.000000	301.000000
mean	2013.627907	4.661296	7.628472	36947.205980	0.043189
std	2.891554	5.082812	8.642584	38886.883882	0.247915
min	2003.000000	0.100000	0.320000	500.000000	0.000000
25%	2012.000000	0.900000	1.200000	15000.000000	0.000000
50%	2014.000000	3.600000	6.400000	32000.000000	0.000000
75%	2016.000000	6.000000	9.900000	48767.000000	0.000000
max	2018.000000	35.000000	92.600000	500000.000000	3.000000

```
df.isnull().sum()
```

	0
Car_Name	0
Year	0
Selling_Price	0
Present_Price	0
Driven_kms	0
Fuel_Type	0
Selling_type	0
Transmission	0
Owner	0

dtype: int64

```
df['Car_Age'] = 2025 - df['Year']
```

```
df.head()
```

	Car_Name	Year	Selling_Price	Present_Price	Driven_kms	Fuel_Type	Selling_type	Transmission	Owner	Car_Age
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0	11
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0	12
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0	8
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0	14
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0	11

```
df.drop(['Car_Name', 'Year'], axis=1, inplace=True)
```

```
df.head()
```

	Selling_Price	Present_Price	Driven_kms	Fuel_Type	Selling_type	Transmission	Owner	Car_Age
0	3.35	5.59	27000	Petrol	Dealer	Manual	0	11
1	4.75	9.54	43000	Diesel	Dealer	Manual	0	12
2	7.25	9.85	6900	Petrol	Dealer	Manual	0	8
3	2.85	4.15	5200	Petrol	Dealer	Manual	0	14
4	4.60	6.87	42450	Diesel	Dealer	Manual	0	11

```
df = pd.get_dummies(df, drop_first=True)
```

```
df.head()
```

	Selling_Price	Present_Price	Driven_kms	Owner	Car_Age	Fuel_Type_Diesel	Fuel_Type_Petrol	Selling_type_Individual	Transmi
0	3.35	5.59	27000	0	11	False	True	False	
1	4.75	9.54	43000	0	12	True	False	False	
2	7.25	9.85	6900	0	8	False	True	False	
3	2.85	4.15	5200	0	14	False	True	False	
4	4.60	6.87	42450	0	11	True	False	False	

```
X = df.drop('Selling_Price', axis=1)
y = df['Selling_Price']
```

```
X.head()
```

	Present_Price	Driven_kms	Owner	Car_Age	Fuel_Type_Diesel	Fuel_Type_Petrol	Selling_type_Individual	Transmission_Manual
0	5.59	27000	0	11	False	True	False	True
1	9.54	43000	0	12	True	False	False	True
2	9.85	6900	0	8	False	True	False	True
3	4.15	5200	0	14	False	True	False	True
4	6.87	42450	0	11	True	False	False	True

```
y.head()
```

	Selling_Price
0	3.35
1	4.75
2	7.25
3	2.85
4	4.60

dtype: float64

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)
```

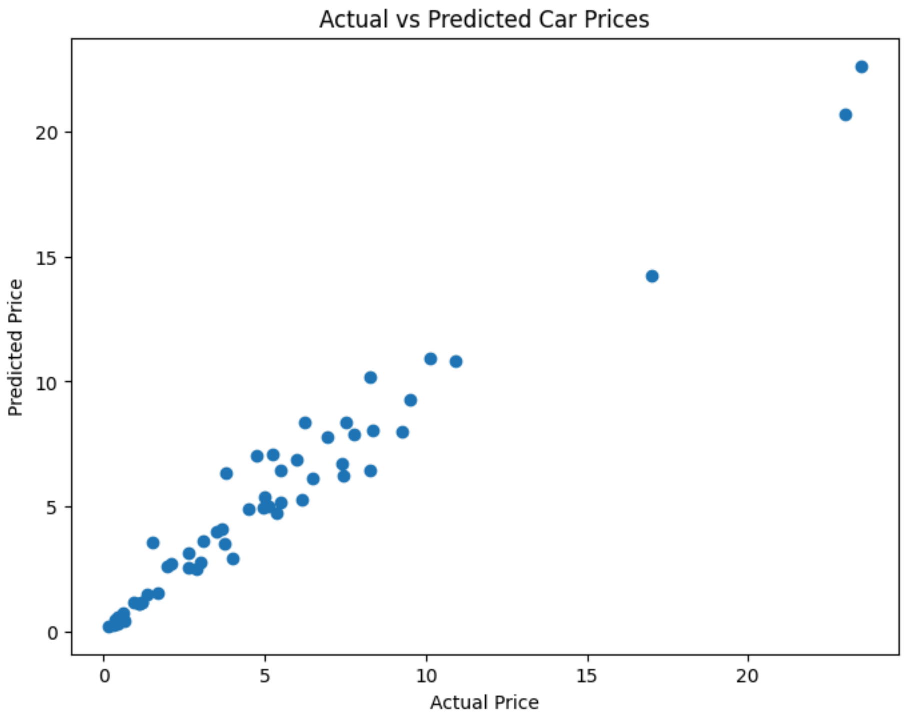
```
▾ RandomForestRegressor
RandomForestRegressor(random_state=42)
```

```
y_pred = model.predict(X_test)
```

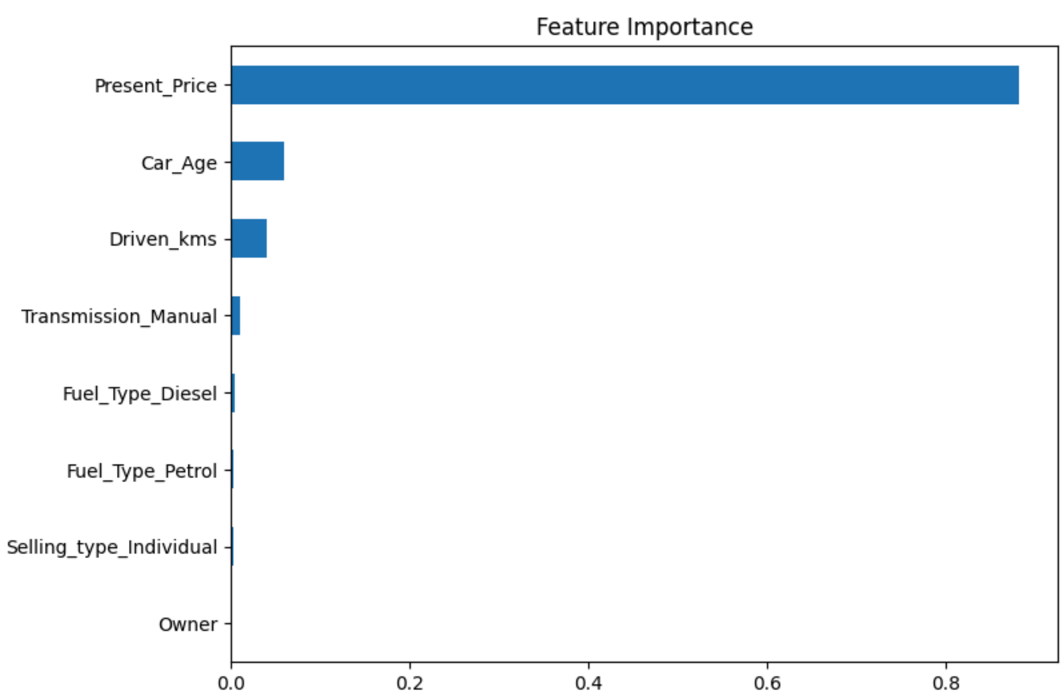
```
from sklearn.metrics import r2_score, mean_absolute_error

print("R2 Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
```

```
R2 Score: 0.9594566919773236
MAE: 0.6368655737704919
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Car Prices")
plt.show()
```




```
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
)

importance.sort_values().plot(
    kind='barh',
    figsize=(8,6)
)

plt.title("Feature Importance")
plt.show()
```




```
from sklearn.metrics import mean_squared_error
import numpy as np

rmse = np.sqrt(mean_squared_error(y_test, y_pred))
```

```
print("RMSE:", rmse)
```